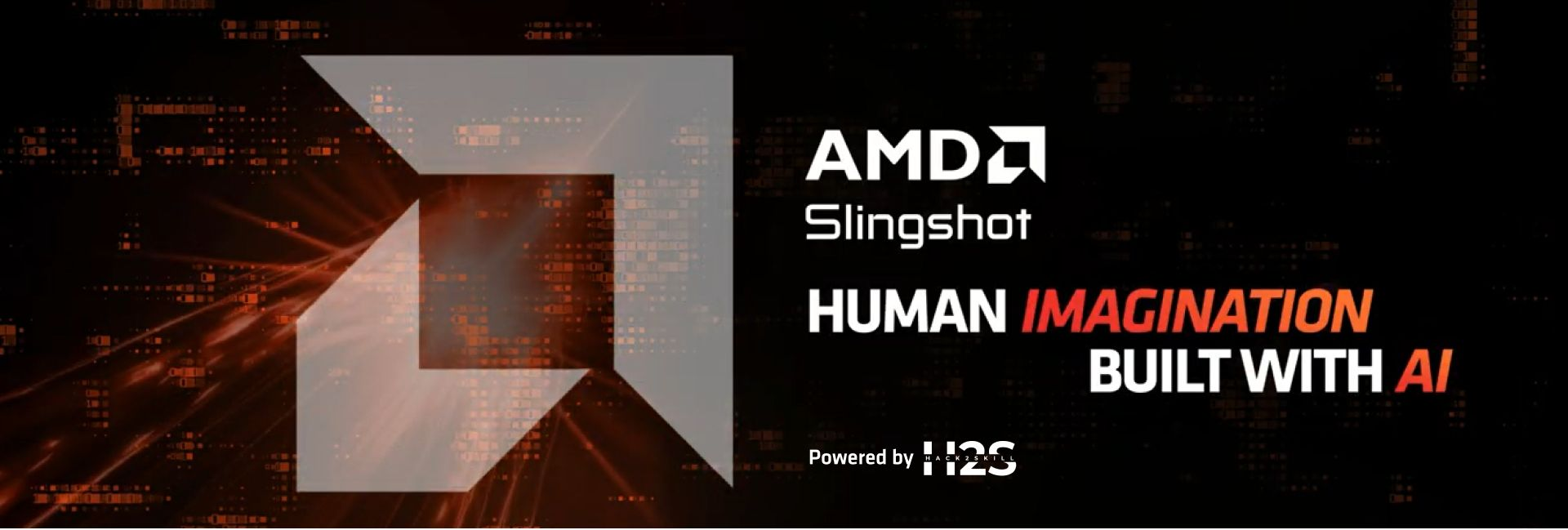


Team Details

- ➔ **Team name -** Vision AI
- ➔ **Team leader name -** Nikhil Kushwaha
- ➔ **Problem Statement -** Large organizations lack intelligent staff tracking across multiple cameras, this project delivers an optimized AI-based surveillance solution.

BRIEF ABOUT THE IDEA

BACKGROUND OF THE PROBLEM DOMAIN

- Monitoring staff presence and movement across multiple areas is increasingly critical in modern organizations.
- Large campuses or multi-room setups make it difficult to track real-time location and movement patterns.
- Manual supervision is inefficient, time-consuming, and prone to human error.
- Lack of integrated analytics prevents understanding of staff mobility, occupancy, and area usage.
- Heterogeneous camera networks introduce challenges in data integration, synchronization, and accurate identification.

NEED FOR AN EFFECTIVE SOLUTION

To provide instant lookup of a staff member's current location across the campus for quicker coordination.

To automatically notify higher authorities if a staff member fails to reach their assigned area within a time threshold (e.g., 10 minutes).

To convert passive camera feeds into actionable situational intelligence that reduces communication delays.

To improve operational responsiveness by supplying real-time location and movement data to managers.

To support incident investigation and space-utilization analytics through stored spatiotemporal logs.

To reduce manual overhead in locating staff and decrease response latency during routine and exceptional situations.

To establish a centralized source of truth for staff distribution, movement trends, and alert history

OPPORTUNITIES

Summary of existing work

- **RFID & Biometric Systems:**
  - Tracks entry/exit only, no live location.
  - Prone to proxy attendance and tailgating.
- **Manual CCTV Monitoring:**
  - Human-dependent, inefficient, and not scalable.
  - No automated alerts or data analytics
- **Basic Facial Recognition:**
  - Works only in controlled, single-point scenarios.
  - Fails in dynamic, multi-camera environments with varying angles and lighting
- **Time-Attendance Software:**
  - Only records timestamps, not physical presence or location.
  - Provides no visibility into intra-day movement or resource utilization.









Limitations of existing solutions

- **No live tracking of person, only basic entry/exit logs.**
- **Cannot dynamically link a person's identity with their current room or zone.**
- **Manual review of hours of footage is extremely time-consuming.**
- **Lack of multi-camera synchronization creates blind spots between zones.**
- **No unified platform to view complete movement history of a specific individual.**
- **Absence of automated alerts for delayed arrival or unauthorized presence.**
- **Systems are mostly reactive, leading to delayed responses.**

USP of proposed solution

- **Need for an Integrated Solution:**
  - A unified system combining real-time face recognition with spatial analytics.
- **Intelligent Automation Deficit:**
  - No existing system automatically alerts managers based on predefined rules (e.g., 10-minute absence).
- **Absence of Movement Pattern Analysis:**
  - A significant gap in solutions that analyze and visualize staff movement patterns for operational efficiency.

LIST OF FEATURES OFFERES BY THE SOLUTION

FEATURES OFFERED

- Accurate real-time identification and tracking of staff across multiple camera zones.
- Centralized dashboard displaying live locations, movement timelines, and alerts.
- Minimal system latency ensuring timely updates and notifications.
- Reliable data generation for behavioral and spatial analysis.
- Demonstrated scalability and adaptability in real-world institutional environments.


Real-time Detection & Location



Attendance & Time Log


Security & Surveillance

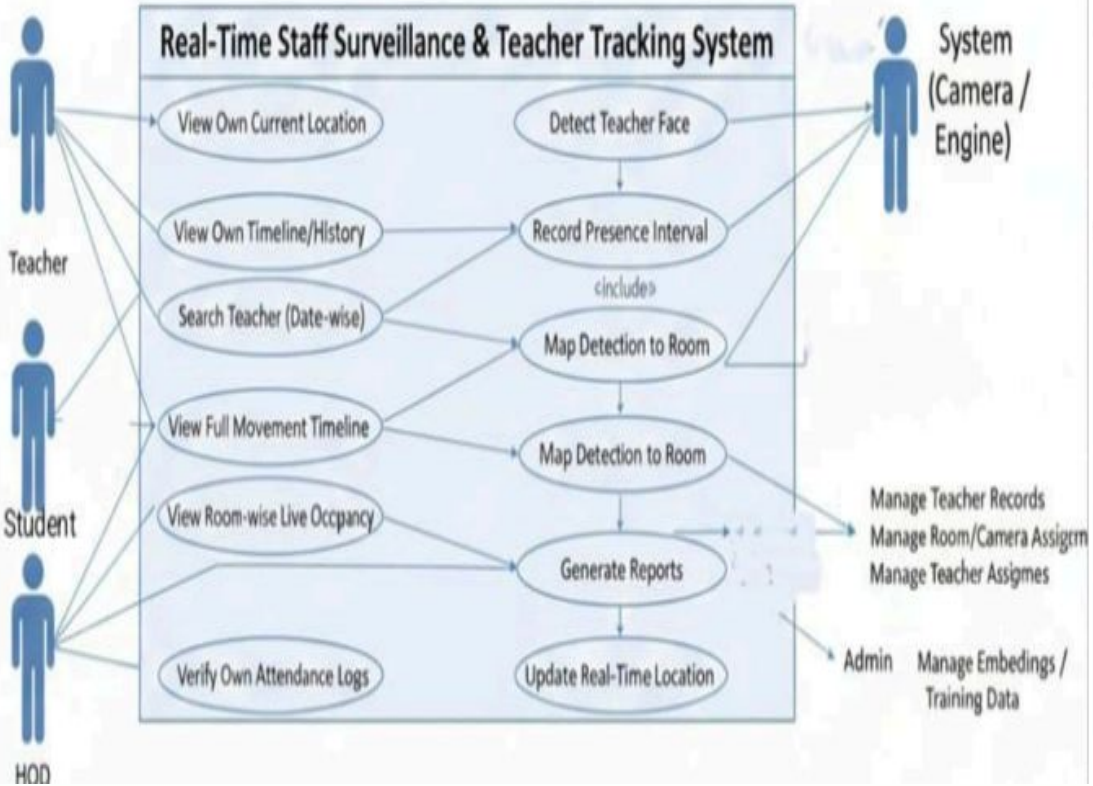


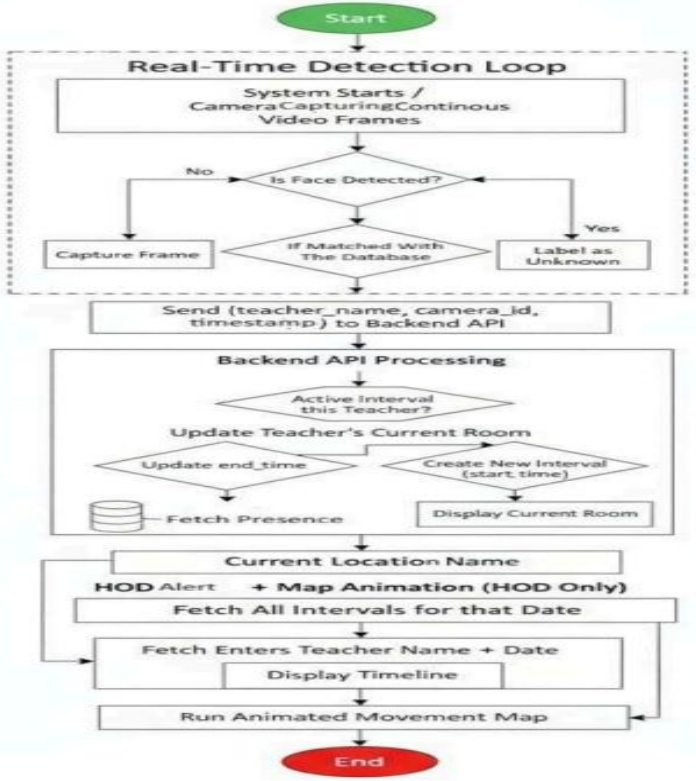
Real-time Availability

BENIFITS

- Enables quick decision-making through real-time visibility of staff presence.
- Enhances operational efficiency and reduces communication delays.
- Improves safety and accountability by monitoring movement patterns.
- Provides actionable analytics for better space and resource management.
- Offers a scalable framework applicable to offices, hospitals, and educational institutions.

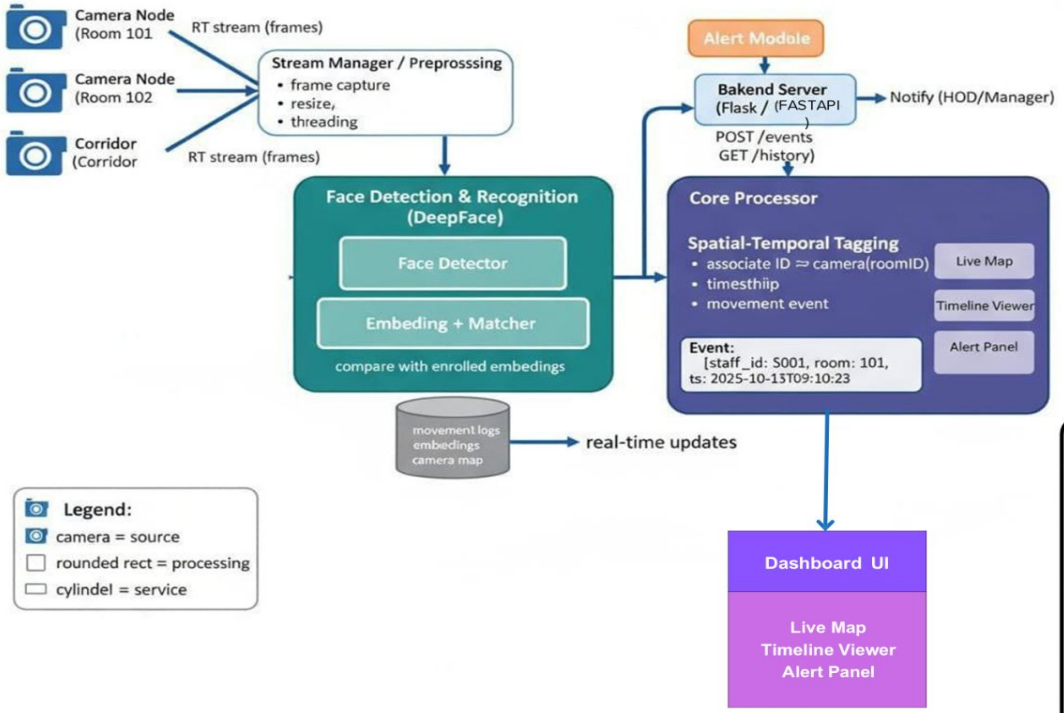
PROCESS-FLOW DIAGRAM AND USE-CASE DIAGRAM





ARCHITECTURE DIAGRAM OF THE POPOSED SOLUTION

HIGH-LEVEL BLOCK DIAGRAM / ARCHITECTURE OVERVIEW



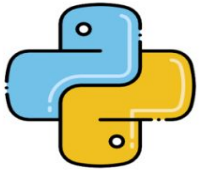
System Modules Planned

- **Image Enrollment Module –** Handles staff registration, face encoding generation, and dataset maintenance.
- **Camera Stream & Detection Module** – Manages multi-camera input, real-time face detection, and frame preprocessing.
- **Recognition & Mapping Module –** Performs identity matching and associates recognized staff with room numbers dynamically.
- **Database & Logging Module –** Stores temporal movement data, recognition history, and camera mappings.
- **Alert & Monitoring Module –** Detects delays or irregular presence and sends automated notifications to management.
- **Visualization Dashboard Module –** Provides live tracking, historical movement analytics, and performance metrics through an intuitive UI

Unique aspects of the approach

- **Integrated Multi-Camera Recognition:** Enables seamless tracking across heterogeneous camera nodes without reinitialization.
- **Real-Time Spatial–Temporal Analytics:** Goes beyond static recognition to map movement patterns and area utilization dynamically.
- **Intelligent Delay Notification System:** Automatically alerts management on delayed presence or irregular staff behavior.
- **Lightweight yet Scalable Architecture:** Optimized for real-time performance even in resource-limited institutional environments.
- **Cross-Domain Adaptability:** The same framework can be extended to corporate, healthcare, or research facilities without modification.

TECHNOLOGY TO BE USED IN THE SOLUTION













Algorithms, Frameworks, and Libraries Planned

Component	Tech / Library	Purpose / Reason
Backend	Python + Flask	Lightweight REST API & HTML rendering. Ideal for rapid prototyping.
Frontend	HTML + JS (vanilla)	Minimal UI for viewing room status & staff timeline. Fetch API for AJAX calls.
Database	SQLite3	Stores staff, cameras, and presence intervals. Simple, file-based DB for prototype.
Face Recognition	DeepFace (Python)	Extracts face embeddings (Facenet model by default) for matching.
Image Capture	OpenCV + requests	Capture frames from IP cameras or phones acting as cameras.
Face Detection	OpenCV Haarcascade	Fast initial detection before embedding computation (reduces DeepFace calls).
Data Serialization	pickle	Stores pre-computed embeddings for fast runtime lookup.
Concurrency	threading	Multi-camera parallel processing in recognizer_intervals_deepface.py.
Dataset Management	Filesystem (dataset/StaffName)	Organizes staff images per person for enrollment.



USAGE OF AMD PRODUCTS/SOLUTIONS

Our AI-based Real-Time Staff Monitoring System requires high computational performance for face detection, feature extraction, and multi-camera processing.

- AMD Ryzen processors support parallel processing of multiple camera streams with multi-core architecture.
- High clock speeds ensure minimal latency during real-time face recognition.
- Efficient handling of deep learning inference tasks such as facial embedding generation.
- Scalable hardware capability suitable for smart campus deployment.
- Energy-efficient processing enables continuous monitoring operations.

The integration of AMD-powered systems enhances the reliability, performance, and scalability of AI-driven institutional monitoring solutions.

Prototype Assets (Optional)

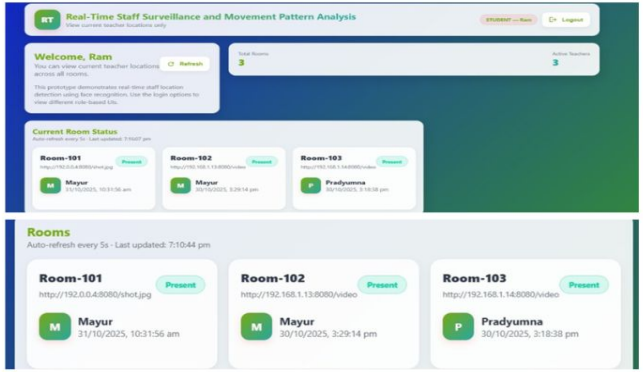
➔ GitHub Public Repository Link-

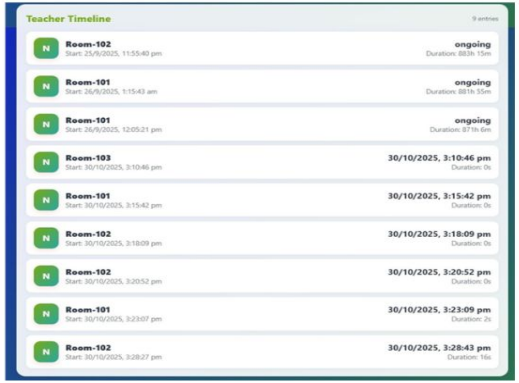
https://github.com/Nikhil139-codes/Staff-tracker.git

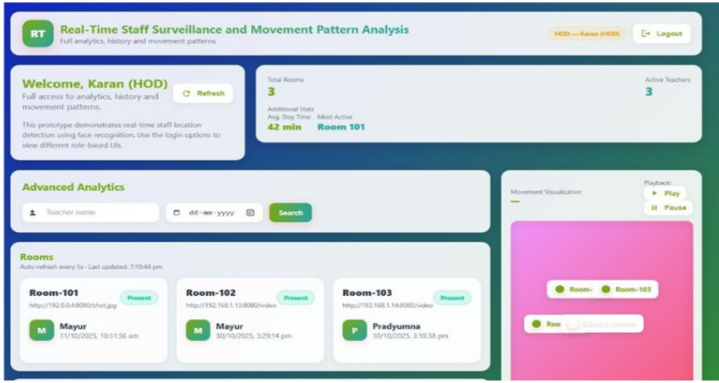
➔ Demo Video Link (Max: 3 Minutes)

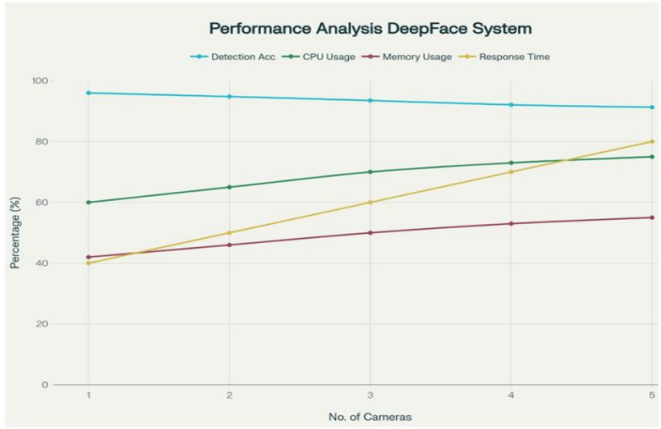
https://youtube.com/shorts/Apfu3lETSp4?si=-wYCJEZSGXJeGQUr

RESULT AND ANALYSIS

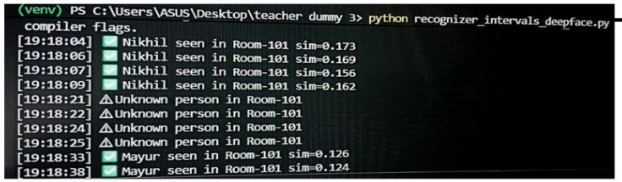


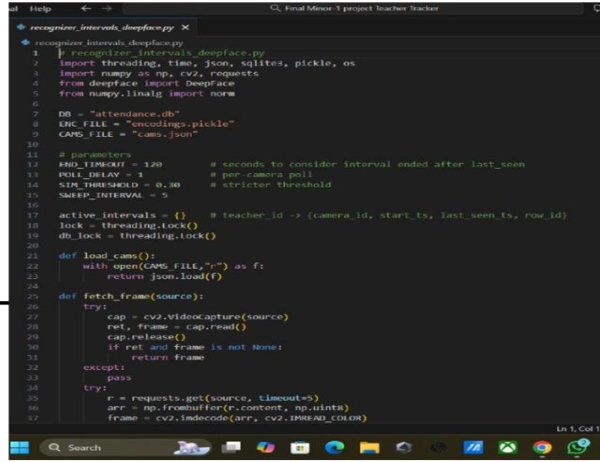












CHALLENGES AND SOLUTION

Technical Challenges

- Handling simultaneous camera streams and ensuring synchronization.
- Maintaining recognition accuracy under varying lighting and angles.
- Reducing latency in real-time face detection and database updates.
- Managing database consistency during parallel thread execution.

Issues faced during Implimentation

- Frequent camera disconnections during testing.
- Delay in UI updates due to high frame processing load.
- Occasional false positives for similar facial features.
- Difficulty in balancing recognition speed and model accuracy.

Adopted Solutions

- Used Python threading for stable multi-camera processing.
- Optimized DeepFace pipeline by pre-filtering frames using Haarcascade.
- Implemented auto-reconnect logic for inactive camera feeds.
- Deployed lightweight Flask APIs with periodic polling for real-time UI refresh.

Lessons Learned and Mitigations

- Efficient data handling and pre-processing significantly improve performance.
- Continuous testing under real environmental conditions enhances reliability.
- Modular design simplifies debugging and future scalability.
- Proper error handling and fail-safe logic are essential for uninterrupted tracking.

FUTURE ENHANCEMENT

Scope for Improvement

Enhance accuracy under varying lighting and crowded conditions.

Implement faster, GPU-accelerated recognition for large-scale deployment.

Improve UI with live floor-map visualization and analytics dashboard.

Next Phase Plans

Integrate mobile notifications for delay or absence alerts.

Add admin panel for staff registration and real-time monitoring control.

Expand system to support cloud storage and remote access.

Long-Term Applications

Smart campus and institutional automation.

Corporate attendance and workforce management systems.

Hospital and research facility staff movement tracking.

Ideas for Further Development

Integrate IoT sensors with facial analytics for energy-efficient room management.

Employ deep-learning models like FaceNet or ArcFace for enhanced precision.

Introduce AI-based anomaly detection for unusual movement or unauthorized presence.


AMD
Slingshot
HUMAN IMAGINATION
BUILT WITH AI
Powered by H2S

Thank you!